

Covariance and Correlation

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Where this fits: Pipeline map step 3 (Understand data). See the [Course map](#).



- **Builds on:** Variance ([Note 222](#)); Descriptive statistics ([Note 230](#)).
- **Leads to:** Correlation significance test ([Note 570](#)).

1. Overview

Key point: Covariance gives the direction of the straight-line relationship between two numerical features; correlation divides it by the two standard deviations, which adds the strength and removes the effect of units.

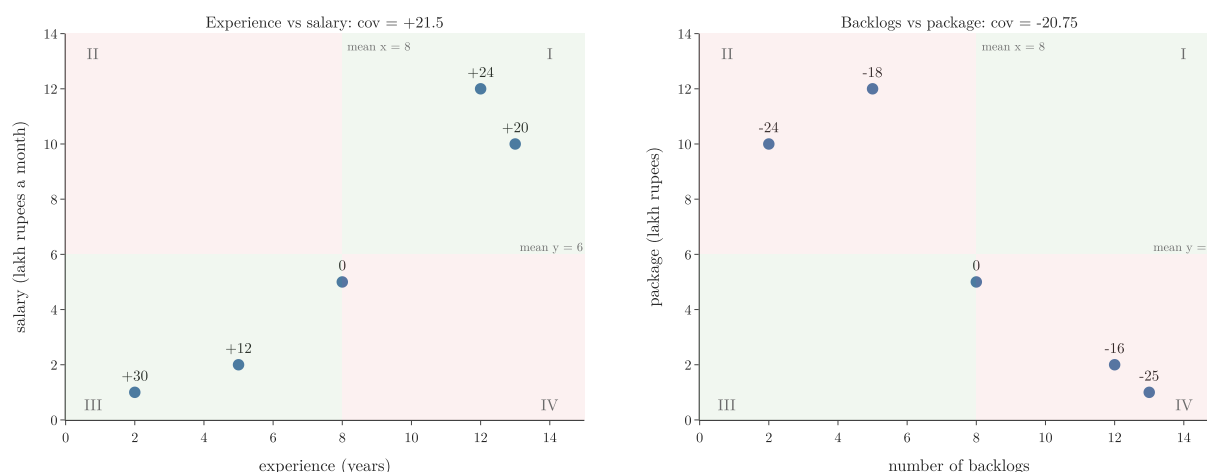


Figure 1: Covariance by quadrants: each point's product of distances from the two means

A **feature** is one variable of the data, one column of the table; an **observation** is one record, one row. A scatter plot shows by eye whether two numerical features rise together. Covariance and correlation turn that picture into a number. Figure 1 shows the idea behind both: the two mean lines cut the plot into four quadrants, and each point votes positive or negative depending on its quadrant.

2. From mean to variance to covariance

Key point: The mean gives the centre, variance gives the spread of one feature, and covariance gives how two features move together.

Each measure fixes a blind spot of the one before:

- **Mean:** gives the centre of a feature, but $-10, 0, 10$ and $-20, 0, 20$ have the same mean, 0.

- **Variance:** gives the spread, and tells those two apart (see the [measures of dispersion Note](#)). But it looks at one feature at a time.
- **Covariance:** the points $(-1, -1), (0, 0), (1, 1)$ rise from left to right, and $(-1, 1), (0, 0), (1, -1)$ fall. Their x and y variances are identical ($2/3$ each), as the [PCA step by step Note](#) (section 3.1) shows. Only a measure that uses both features together can tell them apart.

3. Covariance

Key point: Covariance averages the product of each point's distances from the two means; positive means the features rise together, negative means one falls as the other rises, near zero means no straight-line relationship.

Covariance is taught in the [PCA step by step Note](#) (section 3.2): the average product of each point's distances from the two means, whose sign gives the direction of a linear relationship (see the [bivariate analysis Note](#)). That Note divides by N , as for a whole population; a sample divides by $n - 1$ instead.

$$\sigma_{xy} = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y) \qquad s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

The sample version divides by $n - 1$ for the same reason as the sample variance (see the [measures of dispersion Note](#), section 4.3). For example, five employees (a sample) have experience $x = 2, 5, 8, 12, 13$ years and monthly salary $y = 1, 2, 5, 12, 10$ lakh rupees, with means $\bar{x} = 8$ and $\bar{y} = 6$.

Employee	x	y	$x - \bar{x}$	$y - \bar{y}$	Product
1	2	1	-6	-5	30
2	5	2	-3	-4	12
3	8	5	0	-1	0
4	12	12	4	6	24
5	13	10	5	4	20
Sum					86

$$s_{xy} = \frac{86}{5-1} = 21.5$$

The covariance is positive: more experience goes with a higher salary.

3.1 Reading covariance from the quadrants

Key point: Points in quadrants I and III add to the covariance, points in II and IV subtract from it; whichever side holds more weight sets the sign.

The two mean lines split the scatter plot into four quadrants (Figure 1, left):

- **Quadrant I** (top right): both distances are positive, so the product is positive.
- **Quadrant III** (bottom left): both distances are negative, and negative times negative is positive.
- **Quadrants II and IV:** one distance is positive and one negative, so the product is negative.

So if most points lie in I and III, the covariance is positive; if most lie in II and IV, it is negative. If the points spread evenly over all four, the products cancel and the covariance is near 0. We can often guess the sign just by imagining the two mean lines on a scatter plot.

The right side of Figure 1 shows the opposite case. Five students have 2, 5, 8, 12 and 13 backlogs and packages of 10, 12, 5, 2 and 1 lakh rupees: more backlogs, lower package. With means 8 and 6, the products are $-24, -18, 0, -16, -25$, so

$$s_{xy} = \frac{-24 - 18 + 0 - 16 - 25}{4} = \frac{-83}{4} = -20.75$$

A third case: if every student got the same package of 10 lakh, whatever their backlogs, then $y - \bar{y} = 0$ for every point. Every product is 0, and the covariance is exactly 0: no linear relationship.

3.2 The flaw: covariance depends on the scale

Key point: Multiplying a feature by a number multiplies the covariance by it too, so the size of a covariance says nothing about how strong the relationship is.

Covariance tells the direction of the relationship, but not its **strength**: how closely the points follow a straight line. Its size depends on the units of the features.

1. **In words:** if every x is multiplied by a and every y by c , every distance from the mean is multiplied too, so every product, and the covariance, is multiplied by $a \times c$.
2. **Formula:**

$$\text{cov}(a x, c y) = a c \text{cov}(x, y)$$

3. **Example:** measuring the employees' experience in months instead of years multiplies x by 12:

$$\text{cov} = 12 \times 21.5 = 258$$

Measuring salary in rupees instead of lakhs multiplies the covariance by 100,000: 2,150,000. The employees are the same; only the units changed.

Figure 2 shows the problem with 40 random points. The left panel plots a feature against itself, a perfect straight line, with covariance 1205. The middle panel plots x against a noisier

y , a weaker relationship, with covariance 939. The right panel doubles both features: the picture is identical to the middle one, yet the covariance jumps to 3757, larger than for the perfect line.

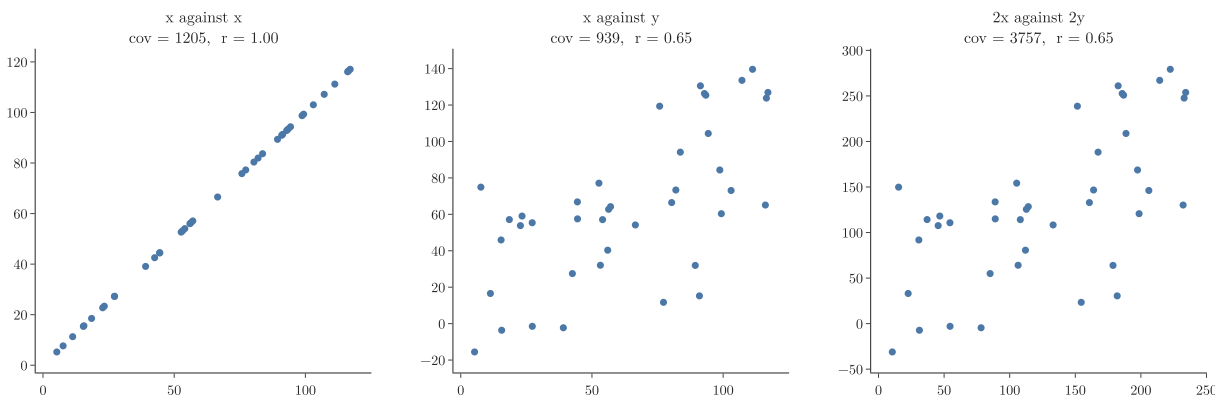


Figure 2: Covariance changes with the scale; correlation does not

So a large covariance does not mean a strong relationship. Covariance is reliable only for its sign. Its main use is as the building block of correlation.

Extra: The covariance of a feature with itself is its variance. With $y = x$ the formula becomes $\sum (x_i - \bar{x})(x_i - \bar{x}) / (n - 1) = \sum (x_i - \bar{x})^2 / (n - 1)$, the sample variance. The same fact explains why the covariance matrix of the [PCA step by step Note](#) has the variances on its diagonal, and why Figure 2's left panel shows the variance of x , 1205.

4. Correlation

Key point: Correlation is covariance divided by the two standard deviations; it always lies between -1 and +1, gives both direction and strength, and does not change with the units.

The Pearson correlation coefficient r appears in the [understanding your data Note](#) (section 9.1). Here we build it from covariance, which shows why it fixes the scale problem.

1. **In words:** divide the covariance by the standard deviation of x and by the standard deviation of y .
2. **Formula:**

$$r = \frac{\text{cov}(x, y)}{s_x s_y}$$

For a population, $\rho = \sigma_{xy} / (\sigma_x \sigma_y)$. The $n - 1$ in the covariance and in the two standard deviations cancel, so both versions give the same number.

3. **Example:** for the five employees, $s_{xy} = 21.5$, $s_x = 4.637$ years and $s_y = 4.848$ lakh, so

$$r = \frac{21.5}{4.637 \times 4.848} = \frac{21.5}{22.48} \approx 0.957$$

For the backlogs data, $r = -20.75 / (4.637 \times 4.848) \approx -0.923$.

Both relationships are strong; one rises and one falls.

4.1 Reading a correlation

Key point: The sign gives the direction and the distance from 0 gives the strength; +1 and -1 are perfect straight lines.

The scale of r , from -1 to $+1$, is read as in the [understanding your data Note](#) (section 9.1): the sign gives the direction, and the closer $|r|$ is to 1, the closer the points lie to a straight line, so the stronger the relationship. Unlike covariance, which can be any number (1205, 3757, -5000), r never leaves that range.

In Figure 2, x against itself gives $r = 1.00$. x against y gives $r = 0.65$: positive, but the points scatter around the line, so it is clearly below 1.

Extra: Rules of thumb put names on r , but fields disagree. One common scale calls $|r|$ of 0.7 to 0.9 strong, 0.4 to 0.6 moderate and 0.1 to 0.3 weak; a medical scale calls 0.7 only moderate (Akoglu 2018, Table 1). On the first scale, the employees' $r = 0.957$ is strong and Figure 2's $r = 0.65$ is moderate.

4.2 Correlation does not depend on the scale

Key point: Scaling a feature scales its standard deviation by the same amount, which cancels the change in the covariance.

In Figure 2, doubling both features quadrupled the covariance but left r at 0.65. Scaling by positive numbers never changes the correlation.

Extra: Proof that correlation ignores the units.

1. **In words:** multiplying x by a multiplies the covariance by a and the standard deviation of x by a too, so the two cancel. The same holds for y . (Adding a constant changes nothing at all, since it moves the mean by the same amount.)
2. **Formula:** for $a, c > 0$,

$$r(ax, cy) = \frac{ac \operatorname{cov}(x, y)}{(a s_x)(c s_y)} = \frac{\operatorname{cov}(x, y)}{s_x s_y} = r(x, y)$$

If a or c is negative, the sign of r flips but its size stays.

3. **Example:** experience in months: $\operatorname{cov} = 258$ and $s_x = 12 \times 4.637 = 55.64$, so $r = 258 / (55.64 \times 4.848) \approx 0.957$, as before.

Because it gives both the direction and the strength, and does not depend on units, correlation is the measure we use to study the linear relationship between two numerical features, for example before linear regression. Covariance is still needed, as the step that computes it.

Python: Covariance and correlation.

```
import numpy as np
```

```
x = np.array([2, 5, 8, 12, 13])
y = np.array([1, 2, 5, 12, 10])
np.cov(x, y)[0, 1]          # 21.5 (divides by  $n - 1$ )
np.corrcoef(x, y)[0, 1]     # 0.957
```

Both return a 2 by 2 matrix; [0, 1] picks the value for the pair. In pandas, `df["a"].cov(df["b"])` and `df["a"].corr(df["b"])`. The Notebook (`notebook.ipynb`) reruns the scale experiment of Figure 2.

5. Correlation does not imply causation

Key point: Two features can move together without one causing the other, often because a hidden third factor drives both.

Causation is a cause-and-effect relationship: a change in one thing produces a change in the other. A correlation only says two things move together. The phrase “correlation does not imply causation” means that a correlation, however strong, is not evidence that one variable causes the other.

Two examples:

- **Firefighters and fire size.** At bigger fires, more firefighters are present: a strong positive correlation. Reading it as “more firefighters make bigger fires” is obviously wrong; the size of the fire decides how many firefighters are sent. Here the direction is clear, but in many datasets it is not.
- **Ice cream and drownings.** On days when more ice cream is sold, more people drown. Ice cream does not cause drowning. Hot weather drives both: when it is hotter, more people buy ice cream and more people go swimming (Figure 3). A third factor of this kind is what statisticians call a confounder (Freedman et al. 2007, ch. 2).

Correlation is not causation

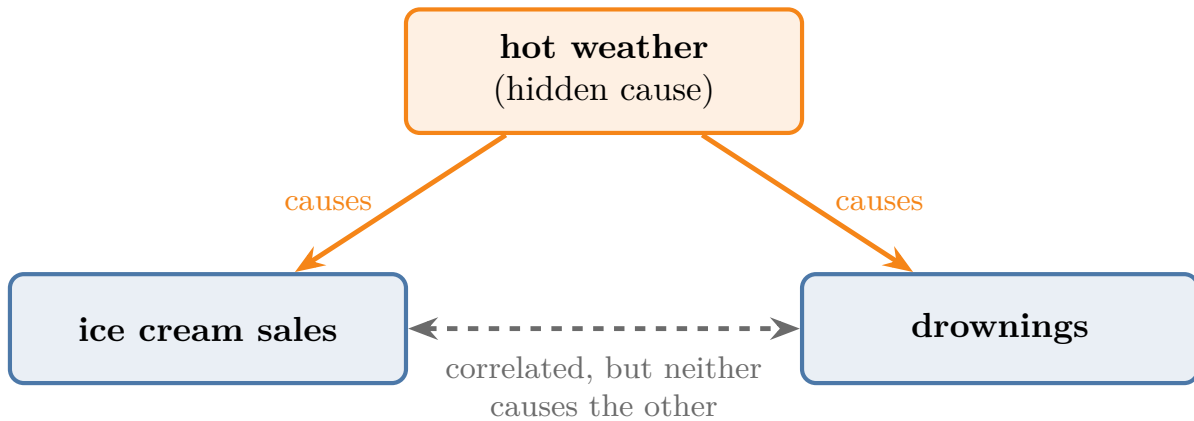


Figure 3: A hidden cause makes two features correlated

A hidden factor that drives two variables, like the weather here, is called a **confounding variable** (or confounder). Experience and salary are a subtler case: they are correlated, but the salary may come from skills that grow with experience, not from the years themselves.

Establishing causation needs more than data that happens to be collected: controlled experiments such as **randomised controlled trials**, where a random half of the subjects gets a treatment and the other half does not, or carefully designed observational studies. Until then, a correlation is a hint worth investigating, not a conclusion.

6. Summary

Measure	Formula	Range	Tells	Depends on units?
Covariance (sample)	$\sum(x_i - \bar{x})(y_i - \bar{y})/(n - 1)$	any number	direction	yes
Correlation r	$\text{cov}(x, y)/(s_x s_y)$	-1 to +1	direction and strength	no

Example	Covariance	r
Experience vs salary	21.5	0.957
Backlogs vs package	-20.75	-0.923
Same package for everyone	0	undefined (no spread in y)

- Points in quadrants I and III push the covariance up; II and IV push it down.
- $\text{cov}(ax, cy) = ac \text{cov}(x, y)$, but r stays the same.
- The covariance of a feature with itself is its variance.
- Correlation does not imply causation; a confounding variable can drive both features.

7. Sources

- Akoglu, H. (2018). User's guide to correlation coefficients. *Turkish Journal of Emergency Medicine*, 18(3), 91-93. Table 1.
- Freedman, D., Pisani, R. and Purves, R. (2007). *Statistics*, 4th ed. W. W. Norton. Chapter 2, Observational Studies (association is not causation; confounding factors).

8. Key terms

Term	Meaning
Feature	One variable of the data, one column of the table
Observation	One record, one row of the table
Population covariance (σ_{xy})	Covariance of a whole population, dividing by N
Sample covariance (s_{xy})	Covariance of a sample, dividing by $n - 1$
Strength of a relationship	How closely the points follow a straight line; measured by $ r $
Causation	A cause-and-effect relationship: changing one thing changes the other
Confounding variable	A hidden factor that drives two variables and makes them correlated

Term	Meaning
Randomised controlled trial	An experiment that assigns a treatment at random, to test causation